Complex Stochastic Amplitudes: A Unified Framework for Interference, Superposition, and Noise in Classical Probability

Pearl Bipin Pulickal

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Abstract

We propose a rigorous mathematical framework for complex-valued probability amplitudes, generalizing classical probability through the use of L^2 spaces. By modeling events using complex amplitudes and defining probabilities as squared magnitudes of integrals, we introduce a clean method of analyzing interference, superposition, and stochastic evolution in probabilistic systems. This paper constructs the theoretical foundation and explores applications to wave interference and noise in dynamic systems.

1 Introduction

Traditional probability theory is based on real-valued measures over sigma-algebras. However, complex-valued functions, particularly in L^2 spaces, offer a more nuanced and expressive way to model systems where interference and phase relationships play a role—e.g., in wave mechanics, quantum physics, or signal processing. This paper builds a framework using complex analysis and probability theory to rigorously define and study such systems.

2 Complex Probability Amplitudes

Let $(\Omega, \mathcal{F}, \mu)$ be a measure space. Define a complex-valued function $\psi : \Omega \to \mathbb{C}$ such that:

$$\int_\Omega |\psi(\omega)|^2 \, d\mu(\omega) = 1 \tag{1}$$

Then $\psi \in L^2(\Omega, \mu)$ is called a probability amplitude.

For any measurable event $A \in \mathcal{F}$, define the probability as:

$$P(A) = \left| \int_A \psi(\omega) \, d\mu(\omega) \right|^2 \tag{2}$$

This differs from classical probability where $P(A) = \int_A p(\omega) d\mu$ for some real-valued density $p(\omega) \geq 0$.

2.1 Example: Two-State System

Let $\Omega = \{\omega_1, \omega_2\}$ with $\mu(\{\omega_1\}) = \mu(\{\omega_2\}) = 1$. Let $\psi(\omega_1) = \frac{1}{\sqrt{2}}, \psi(\omega_2) = \frac{i}{\sqrt{2}}$. Then:

$$P(\{\omega_1\}) = \left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2},$$
$$P(\{\omega_2\}) = \left|\frac{i}{\sqrt{2}}\right|^2 = \frac{1}{2},$$
$$P(\Omega) = \left|\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}\right|^2 = \left|\frac{1+i}{\sqrt{2}}\right|^2 = 1$$

3 Interference and Superposition

Suppose two amplitudes ψ_1 and ψ_2 are given, and define a new amplitude:

$$\psi = \alpha\psi_1 + \beta\psi_2, \quad \text{with } \alpha, \beta \in \mathbb{C}, \quad |\alpha|^2 + |\beta|^2 = 1 \tag{3}$$

Then the probability of event A is:

$$P(A) = \left|\int_A (\alpha\psi_1 + \beta\psi_2)\, d\mu\right|^2 \tag{4}$$

Cross terms in the squared magnitude account for interference:

$$P(A) = |\alpha|^2 P_1(A) + |\beta|^2 P_2(A) + 2\Re\left(\alpha\overline{\beta}\int_A \psi_1\overline{\psi_2}\, d\mu\right) \tag{5}$$

4 Stochastic Evolution of Phase

Let the amplitude evolve over time as:

$$\psi_t(\omega) = \psi_0(\omega)e^{i\phi_t(\omega)} \tag{6}$$

Where $\phi_t(\omega)$ is a real-valued stochastic process (e.g., Brownian motion).

Then:

$$|\psi_t(\omega)|^2 = |\psi_0(\omega)|^2, \quad \Rightarrow \quad \int_\Omega |\psi_t(\omega)|^2\, d\mu = 1 \quad \forall t \tag{7}$$

This models decoherence and random phase shifts in physical or signal systems.

5 Analytic Structure and Complex Integration

Let ψ be holomorphic in a domain $D \subset \mathbb{C}$. The probability over a path $\gamma \subset D$ is:

$$P(\gamma) = \left|\int_\gamma \psi(z)\, dz\right|^2 \tag{8}$$

Using Cauchy's integral theorem or residue calculus, one can evaluate probabilities exactly for special classes of analytic amplitudes.

6 Axiomatic Foundations

Traditional probability satisfies Kolmogorov's axioms: non-negativity, normalization, and countable additivity. Our framework modifies these:

[Modified Probability Axioms] For complex amplitudes $\psi \in L^2(\Omega, \mu)$:

1. **Non-negativity**: $P(A) = \left|\int_A \psi d\mu\right|^2 \geq 0 \ \forall A \in \mathcal{F}$
2. **Normalization**: $P(\Omega) = \|\psi\|^2 = 1$
3. **Countable Interference**: For disjoint $\{A_i\}$,

$$P\left(\bigcup_i A_i\right) = \sum_i P(A_i) + \sum_{i \neq j} \Re\left(\int_{A_i} \psi d\mu \overline{\int_{A_j} \psi d\mu}\right) \tag{9}$$

This explicitly shows violation of classical countable additivity due to interference terms. The structure remains self-consistent through enforcement of the L^2 norm.

7 Joint Systems and Correlation

For product spaces $(\Omega_1 \times \Omega_2, \mathcal{F}_1 \otimes \mathcal{F}_2, \mu_1 \otimes \mu_2)$, define joint amplitudes:

$$\psi(\omega_1, \omega_2) \in L^2(\Omega_1 \times \Omega_2), \quad \|\psi\|^2 = 1 \tag{10}$$

Marginal probabilities emerge via partial integration:

$$P(A_1) = \left|\int_{A_1} \left(\int_{\Omega_2} \psi(\omega_1, \omega_2) d\mu_2(\omega_2)\right) d\mu_1(\omega_1)\right|^2 \tag{11}$$

Correlation between events $A_1 \times A_2$ is captured by:

$$P(A_1 \times A_2) \neq P(A_1)P(A_2) + \text{Interference terms} \tag{12}$$

Example: Entangled States

Let $\Omega = \{0,1\} \times \{0,1\}$ with uniform measure. Define:

$$\psi(\omega_1, \omega_2) = \frac{1}{\sqrt{2}}(\delta_{\omega_1,0}\delta_{\omega_2,0} + \delta_{\omega_1,1}\delta_{\omega_2,1}) \tag{13}$$

Then $P(\{\omega_1 = \omega_2\}) = 1$ with perfect correlation despite independent measures.

8 Measure-Theoretic Considerations

While ψ is a vector in L^2, the induced probability P forms a real measure:

The map $P : \mathcal{F} \to [0, 1]$ defined by $P(A) = |\langle \psi, \chi_A \rangle|^2$ is:

- σ-additive when interference terms cancel (orthogonal events)
- A convex combination of vector measures $|\langle \phi, \chi_A \rangle|^2$ in a doubled Hilbert space

This connects to density matrix formulations where $P(A) = \mathrm{Tr}(\rho \Pi_A)$ for $\rho = |\psi\rangle\langle\psi|$ and projection Π_A.

9 Extended Example: Continuous Wavepacket

Let $\Omega = \mathbb{R}$ with Lebesgue measure. Consider Gaussian amplitude:

$$\psi(x) = \frac{1}{(2\pi\sigma^2)^{1/4}} e^{-x^2/(4\sigma^2)} e^{ikx} \tag{14}$$

Position probability follows $P([a, b]) = \left| \int_a^b |\psi(x)|^2 dx \right|^2 = \mathrm{Erf}(b) - \mathrm{Erf}(a)$ by design. The phase e^{ikx} creates momentum-space interference observable via:

$$\tilde{\psi}(p) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \psi(x) e^{-ipx} dx \tag{15}$$

This demonstrates continuum normalization and phase-dependent interference.

10 Measure-Theoretic Foundations of Complex Amplitudes

Let $(\Omega, \mathcal{F}, \mu)$ be a measure space. The complex amplitude framework induces a nonlinear transformation from $L^2(\Omega, \mu)$ to classical probability measures:

[Induced Real Measure] For any $\psi \in L^2(\Omega, \mu)$, the map $P_\psi : \mathcal{F} \to [0, 1]$ defined by

$$P_\psi(A) = \left| \int_A \psi d\mu \right|^2 \tag{16}$$

is a σ-additive measure when restricted to pairwise orthogonal events: if $\{A_i\}$ are disjoint with $\int_{A_i} \psi d\mu \perp \int_{A_j} \psi d\mu$ for $i \neq j$, then

$$P_\psi \left(\bigcup_i A_i \right) = \sum_i P_\psi(A_i) \tag{17}$$

Proof. For orthogonal events, the cross terms vanish:

$$P_\psi\left(\bigcup_i A_i\right) = \sum_i |\int_{A_i} \psi d\mu|^2 + \sum_{i\neq j} \Re\left(\int_{A_i} \psi d\mu \overline{\int_{A_j} \psi d\mu}\right)$$
$$= \sum_i P_\psi(A_i) \quad \text{by orthogonality}$$

□

This establishes compatibility with classical measure theory for non-interfering events. The general case requires a phase-sensitive Radon-Nikodym derivative:

[Complex Density] For a reference measure μ, define the complex density $\rho : \Omega \to \mathbb{C}$ as

$$\rho(\omega) = \psi(\omega)\overline{\psi(\omega)} = |\psi(\omega)|^2 \tag{18}$$

which recovers classical probability densities when phase factors are uniform.

11 Stochastic Phase Dynamics and Decoherence

Let $\{\phi_t(\omega)\}_{t\geq 0}$ be a Wiener process on $(\Omega, \mathcal{F})$. The choice of Brownian motion is natural for modeling environmental noise:

[Phase Diffusion] For $\psi_t(\omega) = \psi_0(\omega)e^{i\phi_t(\omega)}$ with $\phi_t \sim \mathcal{N}(0, \sigma^2 t)$, the expectation of the amplitude evolves as:

$$\mathbb{E}[\psi_t(\omega)] = \psi_0(\omega)e^{-\sigma^2 t/2} \tag{19}$$

leading to exponential decay of coherence.

Proof. Using characteristic functions of Brownian motion:

$$\mathbb{E}[e^{i\phi_t}] = e^{-\frac{1}{2}\sigma^2 t} \quad \text{(Itô calculus)}$$
$$\Rightarrow \mathbb{E}[\psi_t] = \psi_0 e^{-\sigma^2 t/2}$$

□

This models decoherence in open systems. The time-evolved probability becomes:

$$P_t(A) = \int_A |\psi_0(\omega)|^2 d\mu(\omega) + \iint_{A\times A} |\psi_0(\omega)\psi_0(\omega')|e^{-\sigma^2 t}\cos(\theta_0(\omega) - \theta_0(\omega'))d\mu^2 \tag{20}$$

where θ_0 is the initial phase. The second term vanishes as $t \to \infty$, recovering classical probabilities.

12 Applications to Signal Processing

Consider a quadrature amplitude modulation (QAM) system where symbols are encoded in $\psi \in \mathbb{C}^N$ with $\|\psi\|^2 = 1$. Through noisy channels:

[Phase Noise in Communications] Let received signal be $\psi_{rec} = \psi e^{i\phi} + n$ where ϕ is Brownian phase noise and n additive Gaussian noise. The detection probability for symbol k is:

$$P(k) = |\langle \psi_{rec}, \psi_k \rangle|^2 = e^{-\sigma^2 T} |\langle \psi, \psi_k \rangle|^2 + \text{noise terms} \tag{21}$$

where T is symbol duration. Our framework naturally captures the interplay between phase diffusion and detection errors.

13 Financial Modeling Application

Model market states as complex amplitudes where phase correlations encode arbitrage relationships:

[Arbitrage Amplitude] For assets A, B with prices p_A, p_B, define the relative value amplitude:

$$\psi_{AB} = \frac{1}{\sqrt{2}} \left(\sqrt{\frac{p_A}{p_A + p_B}} + e^{i\theta} \sqrt{\frac{p_B}{p_A + p_B}} \right) \tag{22}$$

where θ encodes market sentiment. Arbitrage exists when $P(\text{mispricing}) = |\psi_{AB} - \psi_{BA}|^2 > \epsilon$.

[No-Free-Lunch] If $\exists$ risk-neutral measure μ where all ψ_{AB} are pairwise orthogonal, then:

$$P(\text{arbitrage}) = 0 \quad \text{(Efficient Market)} \tag{23}$$

14 Analytic Structure in Wave Optics

For coherent states $\psi(z) = e^{-|z|^2/2} \sum_{n=0}^{\infty} \frac{z^n}{\sqrt{n!}} |n\rangle$ with $z \in \mathbb{C}$:

[Interferometric Sensing] The probability of detecting a phase shift ϕ in a Mach-Zehnder interferometer is:

$$P(\phi) = \left| \int_\gamma e^{i\phi(z)} \psi(z) dz \right|^2 = e^{-|\alpha|^2 (1 - \cos\phi)} \quad (z = \alpha e^{i\theta}) \tag{24}$$

showing characteristic coherence oscillations.

15 Clarification of Terminology

While borrowing quantum mechanical language, our framework is strictly classical:

- **Entanglement**: Denotes non-separable correlations in complex amplitudes, not non-local quantum states

- **Decoherence**: Models classical information loss to environment, not wavefunction collapse
- **Superposition**: Linear combination of amplitudes with interference, not literal quantum states

The mathematical structures are identical, but physical interpretation remains classical.

16 Conclusion

We have built a logically consistent and mathematically rigorous extension of classical probability theory using complex-valued amplitudes. The framework not only generalizes conventional probability but naturally incorporates interference, phase evolution, and analytic structure—opening the door for applications in quantum mechanics, signal theory, and beyond.